## 0.1 Interior

**Definition 0.1.1.** Let $(X, \mathcal{T})$ be a Topological Space, and $A \subseteq X$ be a subset.
Define an *interior* of $A$ in $X$ as:

$$\mathrm{int}_X(A) \overset{\text{def}}{=} \bigcup_{\substack{U \subseteq A \\ U \text{ is open}}} U$$

When the Ambient space is obvious, then denote $\mathrm{int}_X(A) = A^\circ$.
Note: Trivially, $A^\circ$ is the largest open set contained in $A$.

**Proposition 0.1.2.** Let $(X, \mathcal{T})$ be a Topological Space, and $A, B, A_\alpha \subseteq X$ $(\alpha \in \Lambda)$.

1. $A \subseteq B \implies A^\circ \subseteq B^\circ$
2. $(A \cup B)^\circ \supseteq A^\circ \cup B^\circ$
3. $(A \cap B)^\circ = A^\circ \cap B^\circ$
4. $\left(\bigcap_{\alpha\in\Lambda} A_\alpha\right)^\circ \subseteq \bigcap_{\alpha\in\Lambda} A_\alpha^\circ$
5. $(A - B)^\circ \subseteq A^\circ - B^\circ$

*Proof.* To show 1),
$x \in A^\circ \iff \exists U \in \mathcal{T}$ s.t. $x \in U \subseteq A \subseteq B \implies x \in B^\circ$.
To show 2), using 1). Since $A, B \subseteq A \cup B$, $A^\circ, B^\circ \subseteq (A \cup B)^\circ$. So is the union.
Similarly to 2), $A \cap B \subseteq A, B$ implies $(A \cap B)^\circ \subseteq A^\circ \cap B^\circ$.
To show inverse inclusion, let $x \in A^\circ \cap B^\circ$. Then, there exist $U, V \in \mathcal{T}$ s.t.

$$x \in U \subseteq A, \quad x \in V \subseteq B.$$

Now, $x \in U \cap V \subseteq A \cap B$. This means $x \in (A \cap B)^\circ$.
Finally, for any $\gamma \in \Lambda$, $\bigcap_{\alpha\in\Lambda} A_\alpha \subseteq A_\gamma \implies \left(\bigcap_{\alpha\in\Lambda} A_\alpha\right)^\circ \subseteq A_\gamma^\circ$. Thus $\left(\bigcap_{\alpha\in\Lambda} A_\alpha\right)^\circ \subseteq \bigcap_{\alpha\in\Lambda} A_\alpha^\circ$.
5) is clear, being $(B^c)^\circ \subseteq (B^\circ)^c$. □

**Proposition 0.1.3.** Let $(X, \mathcal{T})$ be a Topological Space, and $A \subseteq X$ be a subspace. For $B \subseteq A$,

$$\mathrm{int}_X(B) = \mathrm{int}_X(A) \cap \mathrm{int}_A(B) \subseteq \mathrm{int}_A(B).$$

## 0.2 Neighborhood and Limit Point

**Definition 0.2.1.** Let $(X, \mathcal{T})$ be a Topological Space.
For a point $x \in X$, a set $N \subseteq X$ is said to be *neighborhood* of $x$ if

$$\exists U \in \mathcal{T} \text{ s.t. } x \in U \subseteq N$$

Denote $\mathcal{N}_x$ as set of open neighborhoods of $x$.

**Definition 0.2.2.** Let $(X, \mathcal{T})$ be a Topological Space, $A \subseteq X$ be a subset.
A point $x \in X$ is called *limit point* of $A$ if

$$\forall U_x \in \mathcal{N}_x, \ A \cap (U_x \setminus \{x\}) \neq \emptyset.$$

Denote $A' \overset{\text{def}}{=} \{x \in X \mid x \text{ is limit point of } A\}$, and it is called *Derived set*.

**Proposition 0.2.3.** Let $(X, \mathcal{T})$ be a Topological Space, $A, B, A_\alpha \ (\alpha \in \Lambda)$ be subsets.

1. $A \subseteq B \implies A' \subseteq B'$
2. $(A \cup B)' = A' \cup B'$
3. $(A \cap B)' \subseteq A' \cap B'$
4. $\left(\bigcup_{\alpha \in \Lambda} A_\alpha\right)' \supseteq \bigcup_{\alpha \in \Lambda} A'_\alpha$

*Proof.* At first, show 1). Let $x \in A'$. Then,

$$x \in A' \iff \forall U \in \mathcal{N}_x, \ A \cap (U \setminus \{x\}) \neq \emptyset.$$

Now, since $A \subseteq B$, $A \cap (U \setminus \{x\}) \subseteq B \cap (U \setminus \{x\}) \neq \emptyset$. Thus $x \in B'$.
To show 3), $A \cap B \subseteq A, B \implies (A \cap B)' \subseteq A' \cap B'$. clear.
To show 2),

$$\begin{aligned} x \in (A \cup B)' &\iff \forall U \in \mathcal{N}_x, \ (A \cup B) \cap (U \setminus \{x\}) \neq \emptyset \\ &\iff \forall U \in \mathcal{N}_x, \ (A \cap (U \setminus \{x\})) \cup (B \cap (U \setminus \{x\})) \neq \emptyset \\ &\iff x \in A' \cup B'. \end{aligned}$$

4) is clear by 1). □

## 0.3 Closure

**Definition 0.3.1.** Let $(X, \mathcal{T})$ be a Topological Space, and $A \subseteq X$ be a subset.
Define a *closure* of $A$ in $X$ as:

$$cl_X(A) \stackrel{\text{def}}{=} \bigcap_{\substack{A \subseteq C \\ C \text{ closed}}} C.$$

When the Ambient space is apparent, then denote $cl_X(A) = \bar{A}$.
Note: Trivially, $\bar{A}$ is the smallest closed set containing $A$.

**Lemma 0.3.2.** Let $(X, \mathcal{T})$ be a Topological Space, and $A \subseteq X$ be a subset. TFAE:

a) $x \in \bar{A}$

b) For any open neighborhood $U$ of $x$, $A \cap U \neq \emptyset$.

**Lemma 0.3.3.** Let $(X, \mathcal{T})$ be a Topological Space, and $A \subseteq X$ be a subset. Then,

$$\overline{A} = A \cup A'$$

**Proposition 0.3.4.** Let $(X, \mathcal{T})$ be a Topological Space, and $A, B, A_\alpha \subseteq X$ $(\alpha \in \Lambda)$.

1. $A \subseteq B \implies \overline{A} \subseteq \overline{B}$
2. $\overline{A \cup B} = \overline{A} \cup \overline{B}$
3. $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$
4. $\overline{\bigcap_{\alpha \in \Lambda} A_\alpha} \supseteq \bigcap_{\alpha \in \Lambda} \overline{A_\alpha}$
5. $\overline{A^c} = (A^\circ)^c$ and $(\overline{A})^c = (A^c)^\circ$.

## 0.4 Countability Properties

**Proposition 0.4.1.** If $\mathcal{C}$ is a collection of mutually disjoint open sets of $\mathbb{R}$, then $\mathcal{C}$ is countable.

*Proof.* Let $\mathcal{C} = \{U_\alpha \mid \alpha \in \Lambda\}$ be a collection of mutually disjoint open sets.
For any $\alpha \in \Lambda$ and $x_\alpha \in U_\alpha$, there exists an open interval $x_\alpha \in I_\alpha \subseteq U_\alpha$.
By density of Rationals, there is a $p_\alpha \in I_\alpha \cap \mathbb{Q} \subseteq U_\alpha$.
Define a map:

$$f : \mathcal{C} \to \mathbb{Q}; U_\alpha \mapsto p_\alpha.$$

Now, the disjointness gives $f$ is injective. That is, $f : \mathcal{C} \to f[\mathcal{C}] \subseteq \mathbb{Q}$, so $\mathcal{C}$ is countable. $\square$

**Proposition 0.4.2.** Countable intersection of open sets of Co-countable space $(X, \mathcal{T})$ is open.

*Proof.* Let $U_n \in \mathcal{T} \ (n \in \mathbb{N})$ be open sets. Then,

$$\mathbb{R} \setminus \left( \bigcap_{n \in \mathbb{N}} U_n \right) = \mathbb{R} \cap \left( \bigcap_{n \in \mathbb{N}} U_n \right)^c = \mathbb{R} \cap \left( \bigcup_{n \in \mathbb{N}} U_n^c \right) = \bigcup_{n \in \mathbb{N}} \underbrace{(\mathbb{R} \setminus U_n)}_{\text{countable}}$$

Since a countable union of countable sets is countable, that is. $\square$